Adjacent-Motion Spatial OPD: No vIoU Gain Over T

Fixed eight-source diagnostic | HC01-HC04 and V01-V04 | postseal evaluation

Eight historically exposed sources; 48 teacher calls; 9 actual fits; 45 actual optimizer steps; 9 full final-model readouts. Old T is reused exactly. Adding Motion-OPD to T changed mean vIoU from 31.2860% to 30.7075% in HC2 and from 22.4419% to 22.4220% in VidSTG. Spatial IoU-OPD was positive in 1 case(s), negative in 2, with the other 5 cases using exact-F no-update outputs. This is a negative aggregate result on this fixed historical cohort, not evidence of route-wide failure or fresh-source performance.

Complete-tube metrics (percent; four cases per domain)

Arm	HC2 vIoU	HC2 tIoU	HC2 sIoU	VidSTG vIoU	VidSTG tIoU	VidSTG sIoU
Frozen	22.2105	46.4149	47.8950	14.4901	27.7780	48.4576
Projection	27.8513	58.4342	47.8950	22.0165	40.8065	48.4576
Temporal QA (reused)	31.2860	60.4860	47.8950	22.4419	40.9049	48.4576
Spatial Direct	21.8013	46.4149	46.6776	14.2152	27.7780	47.9876
Spatial IoU-OPD	21.9557	46.4149	47.1380	14.4677	27.7780	48.4338
Spatial Motion-OPD	21.9558	46.4149	47.1382	14.4673	27.7780	48.4332
QA + Direct	30.4007	60.4860	46.6776	22.0589	40.9049	47.9876
QA + IoU-OPD	30.7073	60.4860	47.1380	22.4225	40.9049	48.4338
QA + Motion-OPD	30.7075	60.4860	47.1382	22.4220	40.9049	48.4332

Case-level vIoU (percent)

Case	Domain	F	Projection	T	Direct	IoU	Motion	T+Direct	T+IoU	T+Motion
HC01	hc2	16.96	15.42	6.92	16.96	16.96	16.96	6.92	6.92	6.92
HC02	hc2	35.82	39.94	52.26	34.14	34.47	34.47	48.16	48.87	48.87
HC03	hc2	12.68	23.78	23.78	12.68	12.68	12.68	23.78	23.78	23.78
HC04	hc2	23.39	32.26	42.18	23.43	23.71	23.71	42.75	43.27	43.27
V01	vidstg	36.18	49.28	51.73	35.08	36.09	36.09	50.20	51.66	51.65
V02	vidstg	9.47	24.91	24.48	9.47	9.47	9.47	24.48	24.48	24.48
V03	vidstg	6.01	6.89	6.89	6.01	6.01	6.01	6.89	6.89	6.89
V04	vidstg	6.30	6.98	6.67	6.30	6.30	6.30	6.67	6.67	6.67

Case comparison counts

Arm	+	-	Neutral	Unknown	Base
Frozen	0	0	0	0	8
Projection	7	1	0	0	0
Temporal QA (reused)	7	1	0	0	0
Spatial Direct	1	2	5	0	0
Spatial IoU-OPD	1	2	5	0	0
Spatial Motion-OPD	1	2	5	0	0
QA + Direct	1	2	5	0	0
QA + IoU-OPD	1	2	5	0	0
QA + Motion-OPD	1	2	5	0	0

Candidate Motion Evidence and Teacher Controls

Source display geometry and independently tracked Source geometry are separate

Tracked-geometry support (four original observation slots)

Group	Candidates	>=2 slots	All 4	Anchor only	Missing slots
New independent track: all	43	9	7	34	106
New independent track: Source	8	5	4	3	11
New independent track: nonSource	35	4	3	31	95
Historical LK/display: all	43	18	11	25	88
Historical Source display	8	8	8	0	0
Historical nonSource LK	35	10	3	25	88

Source display support / independent Source track support

Case	Displayed original	Tracked Source	Paired slots changed
HC01	4	1	0
HC02	4	4	3
HC03	4	1	0
HC04	4	4	3
V01	4	4	3
V02	4	4	3
V03	4	1	0
V04	4	2	1

Old support uses original Source display boxes and historical nonSource LK geometry. New Source boxes are independently tracked; preparation took 10.47 CPU seconds. Per-candidate failure reasons are in the anonymous geometry file.

New adjacent-track work totals

Group	Attempted	Accepted	Rejected	Not attempted
all_candidates	906	836	70	5560
Source	451	444	7	808
nonSource	455	392	63	4752

Teacher controls: denominator is eight cases per condition

Condition	Same geom.	Changed	Valid	Truncated	Abstain/invalid
base	NA	NA	8	0	0
id_swap	6.000000	NA	8	0	0
color_swap	7.000000	NA	8	0	0
order_swap	6.000000	NA	8	0	0
joint_swap	4.000000	NA	8	0	2
wrong_query	NA	8.000000	8	0	7

Candidate Oracle IoU uses GT-known original-four support only. Missing candidate boxes are zero-filled only on that known-GT denominator; GT-unknown slots stay excluded.

Case	Pool Oracle	Displayed Oracle	Selected	Source
HC01	0.196423	0.196423	0.196423	0.588250
HC02	0.615816	0.615816	0.615816	0.609936
HC03	0.000000	0.000000	0.000000	0.589733
HC04	0.465802	0.465802	0.406160	0.402772
V01	0.701516	0.701516	0.701516	0.667700
V02	0.623032	0.623032	0.086872	0.589977
V03	0.000000	0.000000	0.000000	0.445717
V04	0.194859	0.194859	0.000000	0.202740

Spatial Updates and Verification

Direct, IoU-OPD and Motion-OPD; no temporal refit

The fixed scope uses 1,792 parameters and five declared rounds per actual fit. There are 9 actual fits, 15 exact-F no-update placeholders, 45 declared fit rounds and 45 actual Adam steps. The student barrier records 9 full final-model readouts. Motion-OPD adds the fixed adjacent displacement loss with lambda_motion = 1.0.

Per-case fit movement and optimizer steps

Case	Eligible	D steps	IoU steps	Motion steps	D param delta	IoU param delta	Motion param delta
HC01	no / reuse F	0	0	0	NA	NA	NA
HC02	yes	5	5	5	1.573542	1.329292	1.329279
HC03	no / reuse F	0	0	0	NA	NA	NA
HC04	yes	5	5	5	0.789829	0.850435	0.850388
V01	yes	5	5	5	1.588412	1.750772	1.751629
V02	no / reuse F	0	0	0	NA	NA	NA
V03	no / reuse F	0	0	0	NA	NA	NA
V04	no / reuse F	0	0	0	NA	NA	NA

Motion-OPD loss readback by round

Round	Fits	Mean OPD	Mean motion	Mean total	Updated	Valid pairs
1	3	-0.233971	0.000754	-0.233217	3	212
2	3	-0.157788	0.000749	-0.157039	3	212
3	3	-0.140943	0.000753	-0.140191	3	212
4	3	-0.180770	0.000758	-0.180012	3	212
5	3	-0.186479	0.000762	-0.185718	3	212

V03 prior-control evidence: exact reuse; 0 new fits.

Prior arm	Cases	vIoU	tIoU	dense sIoU
F	1	6.006629	14.042553	48.259917
Projection	1	6.891628	15.670103	48.259917
Source_iou	1	5.761919	14.042553	46.289589
T	1	6.891628	15.670103	48.259917
oldAlternativeID3_iou	1	1.360450	14.042553	10.345365
oldR1ID2_iou	1	8.246636	14.042553	58.040965

V03 locked reference anchors; missing labels remain unknown

Role	Candidate IoU	Paired Source IoU	Difference	GT known/unknown	Missing on known
Source	0.445717	0.445717	0.000000	1/3	0
oldR1ID2	0.158103	0.445717	-0.287614	1/3	0
oldAlternativeID3	0.053349	0.445717	-0.392369	1/3	0

The complete anonymous data preserve all metric and comparison rows, candidate geometry and display mappings, postseal Oracle diagnostics, teacher parsed choices/costs, fit movement and every Motion-OPD round. Raw query/response prose, source paths and hashes, absolute physical frame identifiers, GT coordinates, weights and optimizer tensors are excluded. Historical-source evidence does not establish fresh-source generalization or qualify a production route.